

## Domain & Range

Name: \_\_\_\_\_

Pre-Calculus | Guided notes — we fill these in together

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### Vocabulary

**Domain:** all the \_\_\_\_\_ values ( $x$ ) a function is allowed to use.

**Range:** all the \_\_\_\_\_ values ( $y$ ) the function actually produces.

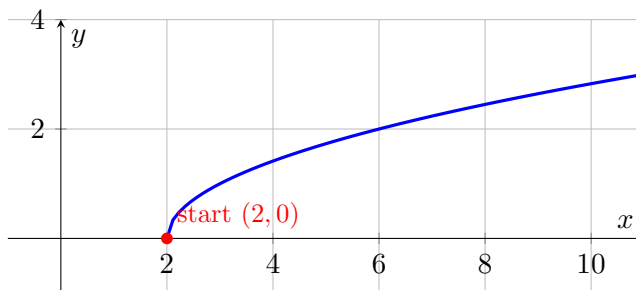
**We do together.** Find the domain and range of  $f(x) = \sqrt{x-2}$ .

**Step 1.** A square root can't hold a negative: set  $x - 2 \geq 0$ , so the domain is \_\_\_\_\_.

**Step 2.** The smallest output is at  $x = 2$ , where  $f(2) = \underline{\hspace{1cm}}$ , then it grows.

**Step 3.** So the range is \_\_\_\_\_.

See it:



$y = \sqrt{x-2}$  — domain  $x \geq 2$ , range  $y \geq 0$ .

**You Try.** Solve each. Show every step, then name the answer.

a) Give the domain of  $f(x) = \frac{1}{x-5}$ .

b) Give the range of  $f(x) = x^2 + 1$ .

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**Summary (fill in):** Watch two traps: a \_\_\_\_\_ can't be 0, and a \_\_\_\_\_ can't hold a negative.